



# Breast Cancer

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Reviewed/Revised Jul 2023 | Modified Apr 2025

**Breast cancers are most often epithelial tumors involving the ducts or lobules. Most patients present with an asymptomatic mass discovered during examination or screening mammography. Diagnosis is made by biopsy. Treatment usually includes surgical excision, often with radiation therapy, and with or without adjuvant chemotherapy, endocrine therapy, or both.**

In the United States, breast cancer is the most common cancer in women, except for skin cancers (1). Breast cancer is the 2nd leading cause of cancer death (the most common is lung cancer) in the overall female population but is the leading cause of cancer death in Black women. Black women are more likely to die from breast cancer than any other race or ethnicity; Asian and Pacific Islander women have the lowest death rate from breast cancer.

In 2023, in women in the United States, it is estimated there will be

- 297,790 new cases of invasive breast cancer
- 55,720 new cases of in situ breast cancer
- 43,700 breast cancer deaths

In 2023, in men in the United States, it is estimated there will be 2800 new cases of invasive breast cancer and 530 deaths from it (2). Manifestations, diagnosis, and management are the same in men and women, although men tend to present at a later stage.

## General references

1. [American Cancer Society](#): Key Statistics for Breast Cancer. Accessed May 4, 2023.
2. [American Cancer Society](#): Key Statistics for Breast Cancer in Men. Accessed May 4, 2023.

## Risk Factors for Breast Cancer

For women in the United States, cumulative lifetime risk of developing breast cancer is about 13% (1 in 8 [1]). Much of the risk is incurred after age 60 (see table [Risk of Being Diagnosed With Invasive Breast Cancer](#)). Risk of dying of breast cancer is about 10% by 5 years after diagnosis (2).

TABLE

**Risk of Being Diagnosed With Invasive Breast Cancer**

| Age (years) | 10-year Risk (%) | 20-year Risk (%) | 30-year Risk (%) | Lifetime Risk (%) | Lifetime Risk of Dying of Invasive Breast Cancer (%) |
|-------------|------------------|------------------|------------------|-------------------|------------------------------------------------------|
| 30          | 0.5              | 2.0              | 4.3              | 13.0              | 2.6                                                  |
| 40          | 1.6              | 3.9              | 7.1              | 12.7              | 2.6                                                  |
| 50          | 2.4              | 5.7              | 9.1              | 11.5              | 2.5                                                  |
| 60          | 3.5              | 7.2              | 9.2              | 9.7               | 2.3                                                  |
| 70          | 4.1              | 6.4              | —                | 7.0               | 1.9                                                  |
| 80          | 3.0              | —                | —                | 3.8               | 1.5                                                  |

Data from 2015–19. Based on [seer.cancer.gov](https://seer.cancer.gov). Accessed April 17, 2023.

Factors that may affect breast cancer risk include the following:

- **Age:** The incidence is highest among women ages 65 to 74 years. Median age at diagnosis is 63 years (1).
- **Race and ethnicity:** The median age at diagnosis is slightly younger for Black women (60 years old) compared to White women (63 years old) (1). Black women have the highest death rate from breast cancer compared with other races or ethnicities; this may be in part because they have a higher rate of triple-negative breast cancer (negative for estrogen and progesterone receptors and human epidermal growth factor receptor [HER2] oncogene), which has a poorer prognosis than other types. White, Asian, and Pacific Islander women are more likely to be diagnosed with localized breast cancer than Black, Hispanic, American Indian, and Alaska Native women.
- **Family history:** Having a 1st-degree relative (mother, sister, daughter) with breast cancer doubles risk of developing the cancer, but breast cancer in more distant relatives increases risk only slightly (3). When  $\geq 2$  1st-degree relatives have breast cancer, risk may be 3 to 4 times higher.
- **Breast cancer gene mutations:** About 5 to 10% of women with breast cancer carry a mutation in one of the two known breast cancer genes, *BRCA1* or *BRCA2*. The risk of developing breast cancer by age 80 is about 72% with a *BRCA1* mutation and about 69% with a *BRCA2* mutation. Women with *BRCA1* mutations also have an approximate 44% lifetime risk of developing ovarian cancer; risk among women with *BRCA2* mutations is

about 17% (4, 5). Men who carry a *BRCA* mutation have a 1 to 2% lifetime risk of developing breast cancer. The mutations are more common among people with Ashkenazi Jewish ancestry. Women with *BRCA1* or *BRCA2* mutations require closer surveillance or preventive measures, such as screening with both mammography and MRI, taking tamoxifen, or undergoing risk-reducing mastectomy.

- **Personal history of breast cancer:** Having had in situ or invasive breast cancer increases risk. Risk of developing cancer in the contralateral breast after mastectomy is about 0.4%/year of follow-up (6).
- **Lobular carcinoma in situ (LCIS):** Having LCIS increases the risk of developing invasive carcinoma in either breast by about 7 to 12 times (7); invasive carcinoma develops in about 1 to 2% of patients with LCIS annually.
- **Gynecologic history:** Early menarche or late menopause increases risk. Risk increases with increasing age at first pregnancy (3).
- **Benign breast disease:** History of a lesion that required a biopsy is associated with a slightly increased risk. Women with multiple breast masses but no histologic confirmation of a high-risk histology should not be considered at high risk. Benign lesions associated with a slightly increased risk of developing invasive breast cancer include complex fibroadenoma, moderate or florid hyperplasia (without atypia), sclerosing adenosis, and papilloma. Risk is about 3 to 5 times higher than average in patients with atypical ductal or lobular hyperplasia (8).
- **Dense breast tissue:** Dense breast tissue seen on screening mammography is associated with a 1.2- to 2.1-fold increased risk of breast cancer (9).
- **Use of oral contraceptives:** Study results vary regarding the use of oral contraceptives and risk of breast cancer. Some studies have found a small increased risk in current or recent users (10).
- **Hormone therapy:** In the Women's Health Initiative randomized trial, menopausal hormone therapy (estrogen plus a progestin) increased risk modestly after only 3 years of use (11). After 5 years of use, the increased risk is about 3 additional cases per 1000 women for each year of use (approximately a 24% increase in relative risk). Use of estrogen alone does not appear to increase risk of breast cancer. Selective estrogen-receptor modulators (eg, raloxifene) reduce the risk of developing breast cancer.
- **Radiation therapy:** Exposure to radiation therapy of the chest up through 45 years old increases risk, with the highest increase for those exposed between ages 10 to 14 years (12). Mantle-field radiation therapy for Hodgkin lymphoma about quadruples risk of breast cancer over the next 20 to 30 years.
- **Diet:** Diet may contribute to development, growth, or prognosis of breast cancers, but conclusive evidence about the effect of a particular diet (eg, one high in fats) is lacking.
- **Obesity:** Postmenopausal women with obesity are at increased risk of breast cancer; studies show a 10% increase in risk for each additional 5 body mass index (BMI) units above normal BMI (13).

- **Smoking and alcohol:** Smoking and alcohol use appear to be associated with an increase in breast cancer risk; the increase with alcohol intake is dose-dependent ([14](#), [15](#)).

The [Breast Cancer Risk Assessment Tool](#) (BCRAT), or Gail model, can be used to calculate a women's 5-year and lifetime risk of developing breast cancer.

## Risk factor references

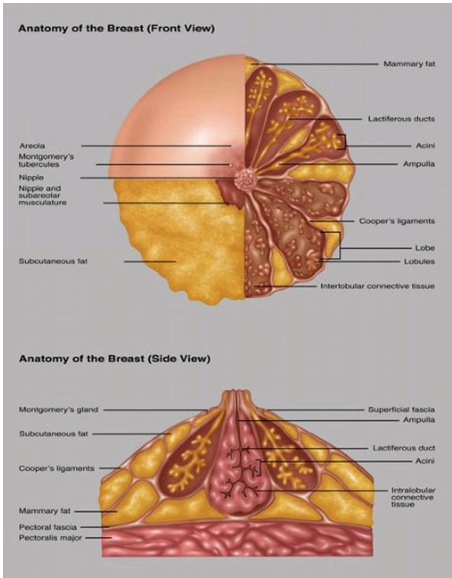
1. [American Cancer Society](#): Key Statistics for Breast Cancer. Accessed May 4, 2023.
2. [American Cancer Society](#): Survival Rates for Breast Cancer. Accessed March 29, 2023.
3. [Collaborative Group on Hormonal Factors in Breast Cancer](#): Familial breast cancer: collaborative reanalysis of individual data from 52 epidemiological studies including 58,209 women with breast cancer and 101,986 women without the disease. *Lancet* 358(9291):1389-1399, 2001. doi:10.1016/S0140-6736(01)06524-2
4. [Kuchenbaecker KB, Hopper JL, Barnes DR, et al](#): Risks of breast, ovarian, and contralateral breast cancer for BRCA1 and BRCA2 mutation carriers. *JAMA* 317 (23):2402–2416, 2017. doi: 10.1001/jama.2017.7112
5. [Breast Cancer Association Consortium; Dorling L, Carvalho S, Allen J, et al](#): Breast cancer risk genes — Association analysis in more than 113,000 women. *N Engl J Med* 4;384 (5):428–439, 2021. doi: 10.1056/NEJMoa1913948
6. [Giannakeas V, Lim DW, Narod SA](#): The risk of contralateral breast cancer: a SEER-based analysis. *Br J Cancer* 125(4):601-610, 2021. doi:10.1038/s41416-021-01417-7
7. [American Cancer Society \(ACOG\)](#): Lobular Carcinoma in Situ (LCIS). Accessed May 4, 2023.
8. [Collins LC, Baer HJ, Tamimi RM, et al](#): Magnitude and laterality of breast cancer risk according to histologic type of atypical hyperplasia: results from the Nurses' Health Study. *Cancer* 109(2):180-187, 2007. doi:10.1002/cncr.22408
9. [American College of Obstetricians and Gynecologists \(ACOG\)](#): Committee opinion no. 625: Management of women with dense breasts diagnosed by mammography. *Obstet Gynecol* 125 (3):750–751, 2015. Reaffirmed 2020. doi: 10.1097/01.AOG.0000461763.77781.79 Accessed May 4, 2023.
10. [American College of Obstetricians and Gynecologists \(ACOG\)](#): Hormonal contraception and risk of breast cancer: Practice advisory. Published 2018, reaffirmed 2022. Accessed May 4, 2023.
11. [Rossouw JE, Anderson GL, Prentice RL, et al](#): Risks and benefits of estrogen plus progestin in healthy postmenopausal women: Principal results from the Women's Health Initiative randomized controlled trial. *JAMA* 288 (3):321–333, 2002. doi:10.1001/jama.288.3.321
12. [John EM, Kelsey JL](#): Radiation and other environmental exposures and breast cancer. *Epidemiol Rev* 15(1):157-162, 1993. doi:10.1093/oxfordjournals.epirev.a036099
13. [Lauby-Secretan B, Scoccianti C, Loomis D, et al](#): Body Fatness and Cancer--Viewpoint of the IARC Working Group. *N Engl J Med* 375(8):794-798, 2016. doi:10.1056/NEJMs1606602
14. [Gram IT, Park SY, Kolonel LN, et al](#): Smoking and Risk of Breast Cancer in a Racially/Ethnically Diverse Population of Mainly Women Who Do Not Drink Alcohol: The MEC Study. *Am J Epidemiol* 182(11):917-925, 2015. doi:10.1093/aje/kwv092
15. [Zhang SM, Lee IM, Manson JE, et al](#): Alcohol consumption and breast cancer risk in the Women's Health Study. *Am J Epidemiol* 165(6):667-676, 2007. doi:10.1093/aje/kwk054

# Pathology of Breast Cancer

Most breast cancers are epithelial tumors (carcinomas) that develop from cells lining ducts or lobules; nonepithelial cancers of the supporting stroma (eg, angiosarcoma, primary stromal sarcomas, phyllodes tumor) are less common.

## Anatomy of the Breast (Front and Side Views)

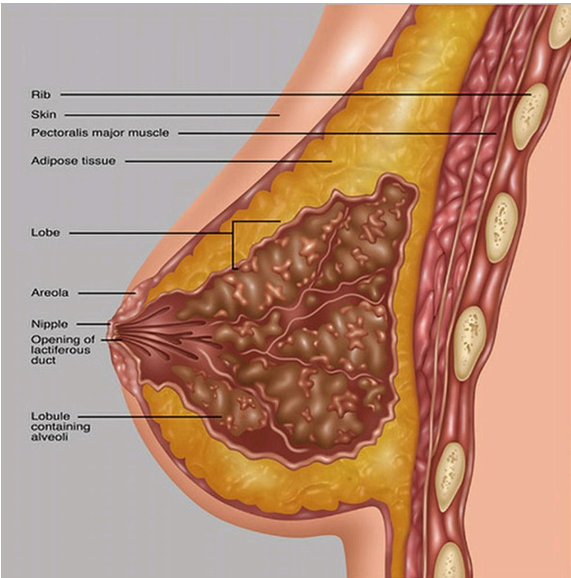
IMAGE



GWEN SHOCKEY/SCIENCE PHOTO LIBRARY

## Anatomy of the Breast (Side View)

IMAGE



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Epithelial cancers are divided into carcinoma in situ and invasive cancer.

**Carcinoma in situ** is proliferation of cancer cells within ducts or lobules but without invasion of stromal tissue. There are 2 types:

- **Ductal carcinoma in situ (DCIS):** DCIS refers to neoplastic epithelial lesions that are confined to the breast ducts; lesions may have varying histology and malignant potential. It is usually detected only by mammography. It may involve a small or wide area of the breast; if a wide area is involved, microscopic invasive foci may develop over time. DCIS accounts for approximately 20% of breast cancers (1). Two randomized studies of DCIS comparing the addition of radiation and tamoxifen to local excision showed a low risk of death from DCIS, regardless of the treatment option chosen (2).
- **Lobular carcinoma in situ (LCIS):** LCIS is often multifocal and is bilateral in approximately 20 to 60% of cases (3). There are 2 types: classic and pleomorphic (cells are larger and more atypical). Classic LCIS is not malignant but increases risk of developing invasive carcinoma in either breast. This nonpalpable lesion is usually suspected based on calcifications or a mass on mammography or ultrasonography and is diagnosed with biopsy. Pleomorphic LCIS behaves more like DCIS; it should be excised to negative margins.

**Paget disease of the nipple** (not to be confused with the metabolic bone disease also called Paget disease) is a form of ductal carcinoma in situ that extends into the skin over the nipple and areola, manifesting with a skin lesion (eg, an eczematous or a psoriaform lesion). Characteristic malignant cells called Paget cells are present in the epidermis. Women with Paget disease of the nipple often have underlying invasive or in situ cancer.

**Invasive carcinoma** is primarily adenocarcinoma. Approximately 75% are infiltrating ductal histology and 10% of the remaining cases are infiltrating lobular (4).

Epithelial cancers may express **hormone receptors** (stromal tumors do not express hormone receptors, eg, phyllodes tumors). Approximately 80% of postmenopausal and 20% of premenopausal patients with breast cancer have an estrogen receptor-positive (ER+) tumor (5); approximately 70% of all breast cancers are progesterone receptor-positive (6). Another cellular receptor is **human epidermal growth factor receptor 2** (HER2; also called HER2/neu or ErbB2); its presence correlates with a poorer prognosis at any given stage of cancer. In approximately 15% of patients with breast cancer, HER2 receptors are overexpressed (7). The majority of breast cancers are hormone receptor-positive and HER2 negative (approximately 70%); 12% are triple negative (hormone receptor-negative and HER2-negative) (7, 8).

**Rare histologic types** that are usually associated with a favorable prognosis include medullary, mucinous, cribriform, and tubular carcinomas (4). Mucinous carcinoma tends to develop in older women and be slow growing.

## Peau d'Orange

IMAGE



Sometimes in inflammatory breast cancer and very advanced cancers, the breast appears inflamed, and the skin appears thickened, resembling orange peel (peau d'orange).

"CREATIVE COMMONS PATIENT WITH INFLAMMATORY BREAST CANCER" BY EPIDEMIOLOGY AND SURGICAL MANAGEMENT OF BREAST CANCER IN GYNECOLOGICAL DEPARTMENT OF DOUALA GENERAL HOSPITAL (SCIENTIFIC FIGURE ON RESEARCHGATE) IS LICENSED UNDER [CREATIVE COMMONS 2.0](#). AVAILABLE FROM: [HTTPS://WWW.RESEARCHGATE.NET/FIGURE/PATIENT-WITH-INFLAMMATORY-BREAST-CANCER\\_FIG2\\_234162338](https://www.researchgate.net/figure/PATIENT-WITH-INFLAMMATORY-BREAST-CANCER_FIG2_234162338) [ACCESSED 3 OCT, 2019].

Rare types that typically have a poor prognosis include metaplastic and inflammatory breast cancer.

**Inflammatory breast cancer** is a fast-growing, particularly aggressive, and often fatal cancer. Cancer cells block the lymphatic vessels in breast skin; as a result, the breast appears inflamed, and the skin appears thickened, resembling orange peel (peau d'orange). Usually, inflammatory breast cancer spreads to the lymph nodes in the armpit. The lymph nodes feel like hard lumps. However, often no mass is felt in the breast itself because this cancer is dispersed throughout the breast.

## Paget Disease of the Nipple

IMAGE



In Paget disease of the nipple, ductal carcinoma in situ extends over the nipple and areola and causes scaling, pruritus, and burning pain in the affected nipple.

DR M.A. ANSARY/SCIENCE PHOTO LIBRARY

## References

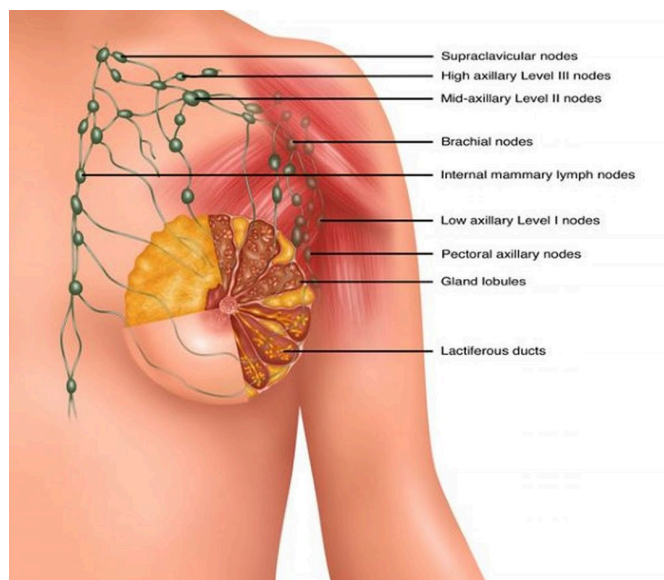
1. [Giaquinto AN, Sung H, Miller KD, et al](#): Breast Cancer Statistics, 2022. *CA Cancer J Clin* 72(6):524-541, 2022. doi:10.3322/caac.21754
2. [Wapnir IL, Dignam JJ, Fisher B, et al](#): Long-term outcomes of invasive ipsilateral breast tumor recurrences after lumpectomy in NSABP B-17 and B-24 randomized clinical trials for DCIS. *J Natl Cancer Inst* 103(6):478-488, 2011. doi:10.1093/jnci/djr027
3. [Wen HY, Brogi E](#): Lobular Carcinoma In Situ. *Surg Pathol Clin* 11(1):123-145, 2018. doi:10.1016/j.path.2017.09.009
4. [American Cancer Society](#): Breast Cancer Facts & Figures. Accessed May 4, 2023
5. [Anderson WF, Chatterjee N, Ershler WB, et al](#): Estrogen receptor breast cancer phenotypes in the Surveillance, Epidemiology, and End Results database. *Breast Cancer Res Treat*. 2002;76(1):27-36. doi:10.1023/a:1020299707510
6. [Li Y, Yang D, Yin X, et al](#): Clinicopathological Characteristics and Breast Cancer-Specific Survival of Patients With Single Hormone Receptor-Positive Breast Cancer. *JAMA Netw Open* 3(1):e1918160, 2020. doi:10.1001/jamanetworkopen.2019.18160
7. [Howlader N, Altekruse SE, Li CI, et al](#): US incidence of breast cancer subtypes defined by joint hormone receptor and HER2 status. *J Natl Cancer Inst* 106(5):dju055, 2014. Published 2014 Apr 28. doi:10.1093/jnci/dju055
8. [National Cancer Institute's Surveillance, Epidemiology, and End Results \(SEER\) Program](#): Cancer Stat Facts: Female Breast Cancer Subtypes. Accessed May 4, 2023

## Pathophysiology of Breast Cancer

Breast cancer invades locally and spreads through the regional lymph nodes, bloodstream, or both. Metastatic breast cancer may affect almost any organ in the body—most commonly, lungs, liver, bone, brain, and skin. Most skin metastases occur near the site of breast surgery; scalp metastases are uncommon.

### Breast Lymph Nodes

IMAGE



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Some breast cancers may recur sooner than others; recurrence can often be predicted based on tumor markers. For example, metastatic breast cancer may occur within 3 years in patients who are negative for tumor markers or occur > 10 years after initial diagnosis and treatment in patients who have an estrogen-receptor positive tumor.

### Hormone receptors

Estrogen and progesterone receptors are nuclear hormone receptors that promote DNA replication and cell division when the appropriate hormones bind to them. Thus, medications that block these receptors are useful in treating and preventing tumors with the receptors.

For tumors in which HER2 receptors are overexpressed, medications that block these receptors are part of standard treatment. HER2 overexpressing tumors respond well to these medications because HER2 is a significant driver of cancer cell progression.

### Breast cancer genes

*BRCA1* and *BRCA2* gene mutations increase the risk of developing breast cancer to 70% (1). Prophylactic bilateral mastectomy reduces the risk of breast cancer by 90% and should be offered to women with a *BRCA* mutation. Other genetic mutations that increase the risk of developing breast cancer include

mutations in *CHEK2*, *PALB2*, *ATM*, *RAD51C*, *RAD51D*, *BARD1*, and *TP53*, which are usually included in panel genetic testing (2).

## Pathophysiology references

1. [Kuchenbaecker KB, Hopper JL, Barnes DR, et al](#): Risks of breast, ovarian, and contralateral breast cancer for BRCA1 and BRCA2 mutation carriers. *JAMA* 317 (23):2402–2416, 2017. doi: 10.1001/jama.2017.7112
2. [Breast Cancer Association Consortium; Dorling L, Carvalho S, Allen J, et al](#): Breast cancer risk genes — Association analysis in more than 113,000 women. *N Engl J Med* 4;384 (5):428–439, 2021. doi: 10.1056/NEJMoa1913948

## Symptoms and Signs of Breast Cancer

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Many breast cancers are discovered as a mass by the patient or during routine physical examination or mammography. Infrequently, the presenting symptom is overall breast enlargement or a nondescript thickening of the breast. Breast pain may be present but is almost never the sole presenting symptom of breast cancer.

A few patients with breast cancer present with signs of metastatic disease (eg, pathologic fracture, abdominal pain, jaundice, dyspnea).

A common finding during physical examination is asymmetry or a dominant mass—a mass distinctly different from the surrounding breast tissue. Diffuse fibrotic changes in a quadrant of the breast, usually the upper outer quadrant, are more characteristic of benign disorders; a slightly firmer thickening in one breast but not the other may be a sign of cancer.

Some types of breast cancer manifest with notable skin changes:

- **Paget disease of the nipple** is associated with an underlying in situ or invasive carcinoma and manifests as skin changes, including erythema, crusting, scaling, and discharge; these changes usually appear so benign that the patient ignores them, delaying diagnosis for a year or more. About 50% of patients with Paget disease of the nipple have a palpable mass at presentation.
- **Inflammatory breast cancer** manifests as erythema and enlargement of the breast, often without a mass, and skin may be discolored or appear thickened, resembling orange peel (peau d'orange). A nipple discharge is common.

**More advanced breast cancers** are characterized by one or more of the following examination findings:

- Fixation of the mass to the chest wall or to overlying skin
- Satellite nodules or ulcers in the skin

Matted or fixed axillary lymph nodes suggest tumor spread, as does supraclavicular or infraclavicular lymphadenopathy.

## Breast Mass

IMAGE



This photo shows an irregular mass on the lateral aspect of the left breast. The mass was later diagnosed as breast cancer.

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## Inflammatory Breast Cancer

IMAGE



This photo shows diffuse erythema and edema in the left breast of patient with inflammatory breast cancer.

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## Diagnosis of Breast Cancer

- Initial detection with mammography and/or other imaging (breast tomosynthesis, ultrasonography, MRI) or clinical breast examination
- Biopsy, including analysis for estrogen and progesterone receptors and HER2 protein

Symptoms, clinical breast examination, or imaging test results may prompt a diagnostic evaluation for breast cancer.

Patients who present with **breast symptoms** (eg, pain, nipple discharge) or a **palpable breast mass** or other abnormality on clinical breast examination are typically evaluated initially with breast ultrasonography. If ultrasound results are abnormal or indeterminate, mammography is done. If imaging findings suggest cancer, biopsy is done.

Even if imaging tests are negative, a biopsy should be performed if a palpable breast mass or other physical findings suggest cancer. If advanced cancer is suspected based on physical examination, biopsy is often done before imaging; however, a prebiopsy bilateral mammogram may help delineate other areas that should be biopsied and provides a baseline for future reference.

For patients with an **abnormal screening mammogram**, image-guided biopsy is performed. An **abnormal breast MRI** is typically evaluated with ultrasonography to determine whether biopsy should be performed under guidance with ultrasonography or MRI.

### Pearls & Pitfalls

- Biopsy should be done if physical findings (eg, palpable mass) suggest breast cancer, even if mammogram results are negative.

## Biopsy

Percutaneous core needle biopsy is preferred to surgical biopsy. Core biopsy can be done guided by imaging or palpation (freehand). Stereotactic biopsy (needle biopsy guided by mammography done in 2 planes and analyzed by computer to produce a 3-dimensional image) or ultrasound-guided biopsy is commonly used to improve accuracy. Clips are placed at the biopsy site so it can be identified with mammography.

If core biopsy is not possible (eg, the lesion is too posterior), surgical biopsy can be done; a guidewire or localizing metal seed is inserted, using imaging for guidance, to help identify the biopsy site.

Any skin taken with the biopsy specimen should be examined because it may show cancer cells in dermal lymphatic vessels.

The excised specimen should be x-rayed, and the x-ray should be compared with the prebiopsy mammogram to determine whether all of the lesion has been removed. If the original lesion contained microcalcifications, mammography is repeated when the breast is no longer tender, usually 6 to 12 weeks after biopsy, to check for residual microcalcifications. If radiation therapy is planned, mammography should be repeated before radiation therapy begins.

## Evaluation after cancer diagnosis

After cancer is diagnosed, a multidisciplinary evaluation is usually done to plan further testing and treatment. The core multidisciplinary team typically includes a breast surgical oncologist, medical oncologist, and radiation oncologist.

A positive biopsy specimen should be analyzed for estrogen and progesterone receptors and for HER2 protein.

### Genetic testing

The National Comprehensive Cancer Network (NCCN) recommends that patients with current or a personal history of breast cancer should be tested for inherited gene mutations that predispose to breast cancer, if they meet any of the following criteria (1):

- Breast cancer diagnosed at age  $\leq 50$  years old
- Treatment indications to aid decisions about use of PARP inhibitors (for metastatic cancer) or olaparib (for high-risk, HER2-negative tumors)
- Triple-negative breast cancer (tumor does not have estrogen or progesterone receptors or overexpression of HER2 protein)
- Multiple primary breast cancers (synchronous or metachronous)
- Lobular breast cancer plus a personal or family history of diffuse gastric cancer
- Male breast cancer
- Ashkenazi Jewish ancestry
- Family history (ie, first-, second-, and third-degree relatives on the same side of the family) with breast cancer at  $\leq 50$  years old, male breast cancer, ovarian cancer, pancreatic cancer, or metastatic or high-risk prostate cancer
- Family history with  $\geq 3$  relatives with either breast or prostate cancer

Some experts have recommended that genetic testing be offered to all patients with breast cancer (2).

For these tests, the best approach is to refer patients to a genetic counselor, who can document a detailed family history, counsel the patient about the risks and benefits of genetic testing, choose the most appropriate tests, and help interpret the results.

### Evaluation for metastatic disease

A complete blood count (CBC) and liver function tests can be done to check for metastatic disease.

An oncologist should determine whether to measure serum carcinoembryonic antigen (CEA), cancer antigen (CA) 15-3, or CA 27-29 and whether bone scanning should be done.

For **bone scanning**, common indications include the following:

- Bone pain
- Elevated serum alkaline phosphatase

- Stage III or IV cancer

**Abdominal CT** is done if patients have any of the following:

- Abnormal liver test results
- Abnormal abdominal or pelvic examination
- Stage III or IV cancer

**Chest CT** is done if patients have either of the following:

- Pulmonary symptoms such as shortness of breath
- Stage III or IV cancer

**MRI** is often used by surgeons for preoperative planning; it can accurately determine tumor size, chest wall involvement, and number of tumors.

## Grading and staging

**Grading** is based on histologic examination of the tissue taken during biopsy. Tumor grade describes how abnormal tumor cells and tissue look under a microscope.

**Staging** follows the TNM (tumor, node, metastasis) classification (see table [Anatomic Staging of Breast Cancer](#)). Because clinical examination and imaging have poor sensitivity for nodal involvement, staging is refined during surgery, when regional lymph nodes can be evaluated. However, if patients have palpably abnormal axillary nodes, preoperative ultrasonography-guided fine needle aspiration or core biopsy may be done:

- If biopsy results are positive, axillary lymph node dissection is typically done during the definitive surgical procedure. However, use of neoadjuvant chemotherapy may make sentinel lymph node biopsy possible if chemotherapy changes node status from N1 to N0. (Results of intraoperative frozen section analysis determine whether axillary lymph node dissection will be needed.)
- If results are negative, a sentinel lymph node biopsy, a less aggressive procedure, may be done instead.

Staging classification follows these models:

- The anatomic staging model, which is based on anatomy of the tumor and which is used in regions of the world where biomarkers cannot be routinely obtained (see table [Anatomic Staging of Breast Cancer](#))
- The prognostic staging model, which is based on anatomy of the tumor as well as status of biomarkers and which is predominantly used in the United States

TABLE

### Anatomic Staging of Breast Cancer\*

| Stage | Tumor | Regional Lymph Node/Distant Metastasis† |
|-------|-------|-----------------------------------------|
| 0     | Tis   | N0/M0                                   |
| IA    | T1‡   | N0/M0                                   |
| IB    | T0    | N1mi/M0                                 |
|       | T1‡   | N1mi/M0                                 |
| IIA   | T0    | N1S/M0                                  |
|       | T1‡   | N1S/M0                                  |
|       | T2    | N0/M0                                   |
| IIB   | T2    | N1/M0                                   |
|       | T3    | N0/M0                                   |
| IIIA  | T0    | N2/M0                                   |
|       | T1‡   | N2/M0                                   |
|       | T2    | N2/M0                                   |
|       | T3    | N1/M0                                   |
|       | T3    | N2/M0                                   |
| IIIB  | T4    | N0/M0                                   |
|       | T4    | N1/M0                                   |
|       | T4    | N2/M0                                   |
| IIIC  | Any T | N3/M0                                   |
| IV    | Any T | Any N/M1                                |

\* Adapted from: Hortobagyi GN, Connolly JL, D'Orsi CJ, Edge SB, Mittendorf EA, et al: Breast. In: Amin MB, Edge S, Greene F, et al, eds; American Joint Committee on Cancer. AJCC cancer staging manual. 8th ed. New York, NY: Springer. 2017. Version 9 changes, when available, can be found on the American Joint Committee on Cancer (AJCC) site [AJCC Version 9 Cancer Staging System](#).

† Classification of regional lymph nodes (N) is divided into clinical (cN) and pathologic (pN), which have slightly different descriptions.

‡ T1 includes T1mi.

§ Here, N1 excludes N1mi.

Tis = ductal carcinoma in situ or Paget disease of the nipple with no tumor (Paget disease with a tumor is classified by tumor size); T1 = tumor  $\leq$  2 cm; T1mi = tumor  $\leq$  0.1 cm; T2 = tumor  $>$  2 but  $\leq$  5 cm; T3 = tumor  $>$  5 cm; T4 = any size tumor with extension to chest wall and/or skin and with ulceration or skin nodules or inflammatory cancer.

NX = Nearby nodes not assessable (for example, because removed previously); N0 = no spread to nearby nodes or only isolated tumor cells; N1 = spread to 1–3 ipsilateral movable, low or midaxillary nodes and/or metastases detected by sentinel lymph node biopsy in clinically negative internal mammary lymph nodes (pN1); N1mi = N1 nodes with micrometastases (about 200 cells  $>$  0.2 mm, but none  $>$  2 mm); N2 = any of the following:

- Spread to ipsilateral low or midaxillary nodes that are fixed or matted, detected by clinical examination (cN2)
- Spread to 4–9 axillary lymph nodes, detected by testing (pN2)
- Spread to internal mammary nodes but not axillary nodes as detected by clinical examination or imaging

N3 = any of the following:

- Spread to ipsilateral internal mammary nodes plus axillary nodes as detected by clinical examination or imaging
- Spread to infraclavicular nodes
- Spread to supraclavicular nodes
- Spread to  $\geq$  10 axillary lymph nodes
- Spread to  $>$  3 axillary lymph nodes and micrometastases or macrometastases by sentinel lymph node biopsy in clinically negative internal mammary nodes (pN3)

M0 = no distant metastases; M1 = distant metastases present.

Adapted from the American Joint Committee on Cancer, *AJCC Cancer Staging Manual, Eighth Edition (2017)*. Springer New York, Inc.

## Fertility preservation

Patients with breast cancer should not become pregnant while being treated for breast cancer. However, all patients who wish to preserve fertility should be referred to a reproductive endocrinologist to discuss fertility preservation before systemic therapy is initiated.

Options for fertility preservation include

- Assisted reproductive techniques (ART) with ovarian stimulation and oocyte or embryo cryopreservation
- Ovarian or testicular tissue cryopreservation

Type of breast cancer, anticipated treatment, and patient preferences affect the type of fertility preservation that can be used. Ovarian suppression (eg, with leuprolide) has been used to minimize the destruction of ova by chemotherapy.

## Diagnosis references

1. [National Comprehensive Cancer Network \(NCCN\)](#): Guidelines, Breast Cancer Version 4.2023.
2. [Manahan ER, Kuerer HM, Sebastian M, et al.](#) Consensus Guidelines on Genetic Testing for Hereditary Breast Cancer from the American Society of Breast Surgeons. *Ann Surg Oncol* 26(10):3025-3031, 2019. doi:10.1245/s10434-019-07549-8

## Treatment of Breast Cancer

- Surgery
- Radiation therapy
- Systemic chemotherapy
- Adjuvant endocrine therapy

For more detailed information about treatment, see [National Comprehensive Cancer Network \(NCCN\) Clinical Practice Guideline: Breast Cancer](#).

For most types of breast cancer, treatment involves surgery and radiation therapy; systemic therapy is given for lymph node involvement or metastatic disease. Choice of treatment depends on tumor and patient characteristics (see table [Treatment by Type of Breast Cancer](#)). Recommendations for surgery are evolving and include early referral to a plastic or reconstruction surgeon for oncoplastic surgery (which combines cancer removal with reconstruction of the breast).

TABLE

**Treatment by Type of Breast Cancer**

| Type                                 | Possible Treatments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DCIS                                 | <p>Mastectomy</p> <p>Breast-conserving surgery in some patients (with lesions confined to one quadrant) with or without* radiation therapy</p> <p>Endocrine therapy for some patients</p>                                                                                                                                                                                                                                                                                                          |
| LCIS, classic                        | <p>Surgical excision to exclude cancer in certain cases</p> <p>If negative, observation with regular examinations and mammograms</p> <p>Tamoxifen or, for some postmenopausal women, raloxifene or aromatase inhibitors to reduce risk of invasive cancer</p> <p>Bilateral prophylactic mastectomy (rarely)</p>                                                                                                                                                                                    |
| LCIS, pleomorphic                    | <p>Surgical excision to negative margins</p> <p>Chemoprevention with tamoxifen or raloxifene for some patients</p>                                                                                                                                                                                                                                                                                                                                                                                 |
| Stages I and II (early-stage) cancer | <p>Preoperative chemotherapy to optimize chances for breast-conserving surgery (in patients with T2 or higher-grade breast cancer)</p> <p>Breast-conserving surgery, followed by radiation therapy</p> <p>Mastectomy with or without breast reconstruction</p> <p>Systemic therapy (eg, postoperative chemotherapy, endocrine therapy, anti-HER2 medications, or a combination) based on results of tumor tests (eg, analysis for hormone receptors and HER2 protein), except possibly in some</p> |

| Type                                                                      | Possible Treatments                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                           | postmenopausal women with hormone receptor-positive tumors with low to intermediate Oncotype Dx scores                                                                                                                                                                                                                                                                                                        |
| Stage III (locally advanced) cancer, including inflammatory breast cancer | <p>Preoperative systemic therapy, usually chemotherapy</p> <p>Breast-conserving surgery or mastectomy if tumor is resectable after preoperative therapy</p> <p>Mastectomy for inflammatory breast cancer</p> <p>Usually, postoperative radiation therapy</p> <p>Postoperative chemotherapy, endocrine therapy, or both</p>                                                                                    |
| Stage IV (metastatic) cancer                                              | <p>If cancer is symptomatic and multifocal, endocrine therapy, ovarian ablation therapy, or chemotherapy</p> <p>If HER2 is overexpressed, anti-HER2 medications (trastuzumab, sometimes with pertuzumab)</p> <p>For brain metastases, local skin recurrences, or isolated symptomatic bone metastases, radiation therapy</p> <p>For bone metastases, IV bisphosphonates to reduce bone loss and bone pain</p> |
| Paget disease of the nipple                                               | <p>Usually, based on type of underlying breast cancer if any</p> <p>Occasionally, local excision only</p>                                                                                                                                                                                                                                                                                                     |
| Locally recurrent breast cancer                                           | <p>Mastectomy or surgical resection (if mastectomy has already been done), sometimes preceded by chemotherapy or endocrine therapy</p> <p>Radiation therapy for some patients</p> <p>Chemotherapy or endocrine therapy</p>                                                                                                                                                                                    |
| Phyllodes tumors                                                          | Wide excision                                                                                                                                                                                                                                                                                                                                                                                                 |

| Type                                                                                                                                                                                                                                                                                                                                                                         | Possible Treatments                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                              | Sometimes radiation therapy (if borderline or malignant)<br>Mastectomy if the mass is large |
| <p>* Wide excision or breast-conserving surgery may be used alone, especially if the lesion is &lt; 2.5 cm and histologic characteristics are favorable, or with radiation therapy if size and histologic characteristics are less favorable.</p> <p>DCIS = ductal carcinoma in situ; HER2 = human epidermal growth factor receptor 2; LCIS = lobular carcinoma in situ.</p> |                                                                                             |

## Surgery

Surgery involves mastectomy or breast-conserving surgery.

**Mastectomy** is removal of the entire breast and includes the following types:

- **Skin-sparing mastectomy:** Spares the pectoral muscles and enough skin to cover the wound, making breast reconstruction much easier, and spares axillary lymph nodes
- **Nipple-sparing mastectomy:** Same as skin-sparing mastectomy plus spares the nipple and areola
- **Simple mastectomy:** Spares the pectoral muscles and axillary lymph nodes
- **Modified radical mastectomy:** Spares the pectoral muscles and removes some axillary lymph nodes
- **Radical mastectomy:** Removes axillary lymph nodes and the pectoral muscles

Radical mastectomy is rarely done unless the cancer has invaded the pectoral muscle.

**Breast-conserving surgery** involves determining the size of the tumor and the required margins (based on the tumor's size relative to the volume of the breast), then surgically removing the tumor with its margins. Various terms (eg, **lumpectomy**, wide excision, quadrantectomy) are used to describe how much breast tissue is removed.

For patients with invasive cancer, survival and recurrence rates with mastectomy do not differ significantly from those with breast-conserving surgery plus radiation therapy as long as the entire tumor can be removed (1).

Thus, patient preference is part of shared decision-making. The main advantage of breast-conserving surgery plus radiation therapy is less extensive surgery and opportunity to keep the breasts. The need for total removal of the tumor with a tumor-free margin overrides any cosmetic considerations. Consulting a plastic surgeon about oncoplastic surgery may help if patients have ptotic (sagging) breasts, while also achieving good resection margins.

Some oncologists treat with neoadjuvant chemotherapy to shrink the tumor before removing it and applying radiation therapy; thus, some patients who might otherwise have required mastectomy can have breast-conserving surgery.

## Lymph node evaluation

During both mastectomy and breast-conserving surgery, axillary lymph nodes are typically evaluated. Methods include

- Axillary lymph node dissection (ALND)
- Sentinel lymph node biopsy (SLNB)

ALND is a fairly extensive procedure that involves removal of as many axillary nodes as possible; adverse effects, particularly lymphedema, are common. Risk of lymphedema is increased for patients with a high preoperative body mass index ( $\text{BMI} \geq 30$ ) and for those with significant weight gain during and after breast cancer treatment (2).

Most surgeons now first do SLNB unless biopsy of clinically suspect nodes detected cancer; risk of lymphedema is less with SLNB. Routine use of ALND is not justified because the main value of lymph node removal is diagnostic, not therapeutic, and SLNB has  $\geq 95\%$  sensitivity for axillary node involvement.

### Lymphedema

IMAGE



This photo shows the arms of a patient with lymphedema of the right (top) arm. Here, it is a result of surgery for breast cancer.

SCIENCE PHOTO LIBRARY

For SLNB, blue dye and/or radioactive colloid is injected around the breast, and a gamma probe (and when dye is used, direct inspection) is used to locate the nodes the tracer drains into. Because these

nodes are the first to receive the tracers, they are considered the most likely to receive any metastatic cells and are thus called sentinel nodes.

If any of the sentinel nodes contain cancer cells, ALND may be necessary, based on numerous factors such as

- Tumor stage
- Hormone receptor status
- Number of involved nodes
- Extranodal extension
- Patient characteristics (3)

Some surgeons do frozen section analysis during mastectomy with SLNB and get prior agreement for ALND in case nodes are positive; others await standard pathology results and do ALND as a 2nd procedure if needed. Frozen section analysis is not routinely done with lumpectomy.

**Impaired lymphatic drainage** of the ipsilateral arm often occurs after axillary node removal (ALND or SLNB) or radiation therapy, sometimes resulting in substantial swelling due to lymphedema. Magnitude of the effect is roughly proportional to the number of nodes removed; thus, SLNB causes less lymphedema than ALND. The lifetime risk of lymphedema after ALND is about 25%. However, even with SLNB, there is a 6% lifetime risk of lymphedema (4). To reduce risk of lymphedema, clinicians usually avoid giving IV infusions on the affected side. Wearing compression garments and preventing infection in the affected limbs (eg, by wearing gloves during yard work) are important. Avoiding ipsilateral blood pressure measurement and venipuncture is sometimes also recommended, even though supporting evidence is minimal (5).

If lymphedema develops, a specially trained therapist must treat it. Special massage techniques used once or twice a day may help drain fluid from congested areas toward functioning lymph basins; low-stretch bandaging is applied immediately after manual drainage, and patients should exercise daily as prescribed. After the lymphedema lessens, typically in 1 to 4 weeks, patients continue daily exercise and overnight bandaging of the affected limb indefinitely.

## Reconstructive procedures

Reconstructive procedures include the following:

- **Prosthetic reconstruction:** Placement of a silicone or saline implant, sometimes after a tissue expander is used
- **Autologous reconstruction:** Muscle flap transfer (using the latissimus dorsi, gluteus maximus, or the lower rectus abdominis) or muscle-free flap transfer

Breast reconstruction can be done during the initial mastectomy or breast-conserving surgery or later as a separate procedure. Timing of surgery depends on patient preference as well as the need for adjuvant therapy such as radiation therapy. However, doing radiation therapy first limits the types of reconstructive surgery that can be done. Thus, consulting a plastic surgeon early during treatment planning is recommended.

Advantages of breast reconstruction include improved mental health in patients who have a mastectomy. Disadvantages include surgical complications and possible long-term adverse effects of implants.

## Contralateral prophylactic mastectomy

Contralateral prophylactic mastectomy is an option for some women with breast cancer (eg, those with a genetic mutation conferring a high risk of breast cancer).

In women with lobular carcinoma in situ in one breast, invasive cancer is equally likely to develop in either breast. Thus, the only way to eliminate the risk of breast cancer for these women is bilateral mastectomy. Some women, particularly those who are at high risk of developing invasive breast cancer, choose this option.

Advantages of contralateral prophylactic mastectomy include

- Decreased risk of contralateral breast cancer (especially in women in a family history of breast or ovarian cancer)
- Improvement in survival in breast cancer patients with an inherited genetic mutation (eg, *BRCA1* or *BRCA2* mutation) and possibly in women diagnosed at age < 50 years
- Decreased anxiety in some patients
- Decreased need for cumbersome follow-up imaging

Disadvantages of contralateral prophylactic mastectomy include

- An almost two-fold increase in surgical complication rates

Contralateral prophylactic mastectomy is not mandatory for patients with the highest risk of developing cancer in the contralateral breast. Close surveillance is a reasonable alternative.

## Radiation therapy

Radiation therapy after breast-conserving surgery significantly reduces incidence of local recurrence in the breast and in regional lymph nodes and may improve overall survival. However, if patients are > 70 and have early ER+ breast cancer, adding radiation therapy to lumpectomy plus tamoxifen may not be necessary; adding radiation therapy does not significantly decrease the rate of mastectomy for local recurrence or the occurrence of distant metastases nor increase the survival rate (6).

Radiation therapy is indicated after mastectomy if any of the following is present:

- The primary tumor is  $\geq 5$  cm.
- Axillary nodes are involved.
- Margins are positive for cancer in resected tissue.

In such cases, radiation therapy after mastectomy significantly reduces incidence of local recurrence on the chest wall and in regional lymph nodes and improves overall survival.

Adverse effects of radiation therapy (eg, fatigue, skin changes) are usually transient and mild. Late adverse effects (eg, lymphedema, brachial plexopathy, radiation pneumonitis, rib damage, secondary cancers, cardiac toxicity) are less common.

To improve radiation therapy, researchers are studying several new procedures. Many of these procedures aim to target radiation to the cancer more precisely and spare the rest of the breast from the effects of radiation.

Adjuvant systemic therapy

(See also [National Comprehensive Cancer Network \(NCCN\) Clinical Practice Guideline: Breast Cancer.](#))

Endocrine therapy, chemotherapy, or HER2-directed therapy delays or prevents recurrence in almost all patients and prolongs survival in some. Hormone positive cancer can be evaluated with Oncotype Dx, which allows targeting chemotherapy and endocrine therapy to the populations most likely to benefit. Hormone negative cancer and HER2 positive cancer are treated with chemotherapy and targeted therapy.

Decisions about the type of adjuvant therapy are made based on tumor characteristics, including estrogen receptor (ER) and progesterone receptor (PR) status, presence of human epidermal growth factor 2 (HER2) protein, grade and stage (including lymph node involvement), and genomic risk stratification. Some patients are treated with a combination of endocrine therapy and chemotherapy.

Chemotherapy is usually begun soon after surgery. If systemic chemotherapy is not required, endocrine therapy is usually begun soon after surgery and is continued for 5 to 10 years.

In higher stage disease, neoadjuvant chemotherapy may be started before surgery.

| TABLE                                              |                                                                                                |                                                                       |
|----------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Preferred Breast Cancer Adjuvant Systemic Therapy* |                                                                                                |                                                                       |
| Axillary Lymph Node                                | ER+ and/or PR+                                                                                 | ER- and PR-                                                           |
| Premenopausal                                      |                                                                                                |                                                                       |
| Negative†                                          | Tamoxifen with or without chemotherapy<br><br>Trastuzumab if HER2+ (sometimes with pertuzumab) | Chemotherapy<br><br>Trastuzumab if HER2+ (sometimes with pertuzumab ) |
| Positive                                           | Chemotherapy plus tamoxifen<br><br>Trastuzumab plus pertuzumab if HER2+                        | Chemotherapy<br><br>Trastuzumab plus pertuzumab if HER2+              |

| Axillary Lymph Node                                                                                                                                                                                                                                                                                 | ER+ and/or PR+                                                                                                                       | ER- and PR-                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Postmenopausal                                                                                                                                                                                                                                                                                      |                                                                                                                                      |                                                                   |
| Negative†                                                                                                                                                                                                                                                                                           | An aromatase inhibitor or tamoxifen (or raloxifene) with or without chemotherapy<br>Trastuzumab if HER2+ (sometimes with pertuzumab) | Chemotherapy<br>Trastuzumab if HER2+ (sometimes with pertuzumab ) |
| Positive                                                                                                                                                                                                                                                                                            | Chemotherapy plus an aromatase inhibitor or tamoxifen (or raloxifene)<br>Trastuzumab plus pertuzumab if HER2+                        | Chemotherapy<br>Trastuzumab plus pertuzumab if HER2+              |
| <p>* For all protocols involving chemotherapy, enrollment in a clinical trial is often considered.</p> <p>† Treatment of node-negative tumors also depends on tumor size and grade.</p> <p>ER = estrogen receptor; HER2 = human epidermal growth factor receptor 2; PR = progesterone receptor.</p> |                                                                                                                                      |                                                                   |

## Endocrine therapy

In patients with ER+ tumors, particularly low-risk tumors, endocrine therapy is typically given. Predictive genomic testing of the primary breast cancer is used to stratify risk in patients and to determine whether combination chemotherapy or endocrine therapy alone is indicated. Common prognostic tests include

- The 21-gene recurrence score assay (based on Oncotype Dx)
- The Amsterdam 70-gene profile (MammaPrint)
- The 50-gene risk of recurrence score (PAM50 assay)

In the United States, most women with breast cancer have ER+/PR+/HER2- breast cancer with negative axillary nodes. In these women, a low or intermediate score on the 21-gene recurrence score assay predicts similar survival rates with chemotherapy plus endocrine therapy and with endocrine therapy alone. Therefore, in this subset of women with breast cancer, chemotherapy may not be necessary.

With endocrine therapy (eg, tamoxifen, aromatase inhibitors), benefit depends on estrogen and progesterone receptor expression; benefit is greatest with the strongest level of hormone receptor expression (7). Medications used for endocrine therapy include tamoxifen and aromatase inhibitors.

**Tamoxifen** is a selective estrogen receptor modulator that competitively binds with estrogen receptors. Adjuvant tamoxifen reduces breast cancer mortality for women with tumors that have estrogen receptors. Tamoxifen therapy for 5 years reduces annual odds of death by about 25% in premenopausal and postmenopausal women regardless of axillary lymph node involvement (8). Treatment for 10 years appears to be even more effective; it prolongs survival and reduces recurrence risk compared with 5 years of treatment (15-year recurrence reduced to 21% from 25%) (9). Tamoxifen can induce or exacerbate menopausal symptoms but reduces incidence of contralateral breast cancer and lowers serum cholesterol. Tamoxifen increases bone density in postmenopausal women and may reduce risk of fractures and ischemic heart disease. However, it significantly increases risk of developing **endometrial cancer**; reported incidence is 1% in postmenopausal women after 5 years of use. Thus, if such women have spotting or bleeding, they must be evaluated for endometrial cancer. Nonetheless, the improved survival for women with breast cancer far outweighs increased risk of death due to endometrial cancer. Risk of thromboembolism is also increased.

**Aromatase inhibitors** (anastrozole, exemestane, letrozole) block peripheral production of estrogen in postmenopausal women. More effective than tamoxifen, these medications are becoming the preferred treatment for early-stage hormone receptor-positive cancer in postmenopausal patients. Letrozole may be used in postmenopausal women who have completed tamoxifen treatment. Optimal duration of aromatase inhibitor therapy is uncertain. A recent trial showed that extending treatment to 10 years resulted in a lower rate of breast cancer recurrence and higher rate of disease-free survival. There was no change in overall survival and a higher rate of fractures and osteoporosis in patients treated for an extended time.

Patients with DCIS are often treated with daily oral tamoxifen.

Preferred therapy for premenopausal women with ER+ breast cancer is typically tamoxifen or ovarian suppression (usually with leuprolide, a GnRH agonist) combined with an aromatase inhibitor. For postmenopausal women, an aromatase inhibitor is preferred. CDK 4/6 inhibitors, such as palbociclib, ribociclib and abemaciclib, are also used for patients with hormone receptor-positive, HER2-negative cancer. Olaparib is used in patients with germline BRCA1 and 2 mutations with HER2-negative breast cancer.

## **Adjuvant chemotherapy**

Usual indications for chemotherapy are one or more of the following:

- Estrogen receptor (ER) and progesterone receptor (PR)-negative
- Human epidermal growth factor 2 (HER2) oncogene-positive (chemotherapy and HER2-directed therapy is given)
- ER/PR+ and positive lymph nodes in a premenopausal patient
- ER/PR+ and HER2- with high Oncotype Dx score

However, studies have shown that chemotherapy is not necessary for many small (< 0.5 to 1 cm) tumors with no lymph node involvement (particularly in postmenopausal patients) because the prognosis is already excellent (10).

Postmenopausal patients with ER– tumors benefit the most from adjuvant chemotherapy (see table [Preferred Breast Cancer Adjuvant Systemic Therapy](#)).

Combination chemotherapy regimens are more effective than a single medication. Dose-dense regimens given for 4 to 6 months are preferred; in dose-dense regimens, the time between doses is shorter than that in standard-dose regimens. There are many regimens; a commonly used one is ACT (doxorubicin plus cyclophosphamide followed by paclitaxel). Acute adverse effects depend on the regimen but usually include nausea, vomiting, mucositis, fatigue, alopecia, myelosuppression, cardiotoxicity, and thrombocytopenia. Growth factors that stimulate bone marrow (eg, filgrastim, pegfilgrastim) are commonly used to reduce risk of fever and infection due to chemotherapy. Long-term adverse effects are infrequent with most regimens; death due to infection or bleeding is rare (< 0.2%).

## **HER2-directed therapy**

If the tumor overexpresses HER2 (HER2+), anti-HER2 monoclonal antibodies (trastuzumab, pertuzumab) may be used. Adding the humanized monoclonal antibody trastuzumab to chemotherapy provides substantial benefit. Trastuzumab is usually continued for a year, although the optimal duration of therapy is unknown. If lymph nodes are involved, adding pertuzumab to trastuzumab improves disease-free survival. A serious potential adverse effect of both these anti-HER2 medications is a decreased cardiac ejection fraction.

## **Metastatic disease**

Any indication of metastases should prompt immediate evaluation. Treatment of metastases increases median survival by 6 months or longer. These treatments (eg, chemotherapy), although relatively toxic, may palliate symptoms and improve quality of life. Thus, the decision to be treated may be highly personal.

Choice of therapy depends on the following:

- Hormone-receptor status of the tumor
- Length of the disease-free interval (from remission to manifestation of metastases)
- Number of metastatic sites and organs affected
- Patient's menopausal status

Systemic endocrine therapy or chemotherapy is usually used to treat symptomatic metastatic disease. Initially, patients with multiple metastatic sites outside the central nervous system (CNS) should be given systemic therapy. If metastases are asymptomatic, there is no proof that treatment substantially increases survival, and it may reduce quality of life.

**Endocrine therapy** is preferred over chemotherapy for patients with any of the following:

- ER+ tumors
- A disease-free interval of > 2 years
- Disease that is not immediately life threatening

In premenopausal women, tamoxifen is often used first. Reasonable alternatives include ovarian ablation by surgery, radiation therapy, and use of a luteinizing-releasing hormone agonist (eg, buserelin, goserelin, leuprolide). Some experts combine ovarian ablation with tamoxifen or an aromatase inhibitor. In postmenopausal women, aromatase inhibitors are being increasingly used as primary endocrine therapy. If the cancer initially responds to endocrine therapy but progresses months or years later, additional forms of endocrine therapy (eg, progestins, the antiestrogen fulvestrant) may be used sequentially until no further response occurs.

The most effective **chemotherapy agents** are capecitabine, doxorubicin (including its liposomal formulation), gemcitabine, the taxanes paclitaxel and docetaxel, and vinorelbine. Response rate to a combination of agents is higher than that to a single agent, but survival is not improved and toxicity is increased. Thus, some oncologists use single agents sequentially.

**Anti-HER2 monoclonal antibodies** (eg, trastuzumab, pertuzumab) are used to treat tumors that overexpress HER2. These are effective in treating and controlling visceral metastatic sites. Trastuzumab is used alone or with endocrine therapy, chemotherapy, or pertuzumab. Trastuzumab plus chemotherapy plus pertuzumab slows the growth of HER2+ metastatic breast cancer and increases survival more than trastuzumab plus chemotherapy (11).

**Tyrosine kinase inhibitors** (eg, lapatinib, neratinib) are being increasingly used in women with HER2+ tumors.

**Radiation therapy** alone may be used to treat isolated, symptomatic bone lesions or local skin recurrences not amenable to surgical resection. Radiation therapy is the most effective treatment for brain metastases, occasionally providing long-term control.

**Palliative mastectomy** is sometimes an option for patients with stable metastatic breast cancer.

**IV bisphosphonates** (eg, pamidronate, zoledronate) decrease bone pain and bone loss and prevent or delay skeletal complications due to bone metastases. About 10% of patients with bone metastases eventually develop hypercalcemia, which can also be treated with IV bisphosphonates.

## Treatment references

1. [Fisher B, Anderson S, Bryant J, et al](#). Twenty-year follow-up of a randomized trial comparing total mastectomy, lumpectomy, and lumpectomy plus irradiation for the treatment of invasive breast cancer. *N Engl J Med* 347(16):1233-1241, 2002. doi:10.1056/NEJMoa022152
2. [Jammallo LS, Miller CL, Singer M, et al](#): Impact of body mass index and weight fluctuation on lymphedema risk in patients treated for breast cancer. *Breast Cancer Res Treat* 142 (1):59–67, 2013. doi: 10.1007/s10549-013-2715-7
3. [Giuliano AE, Hunt KK, Ballman KV, et al](#): Axillary dissection vs no axillary dissection in women with invasive breast cancer and sentinel node metastasis: A randomized clinical trial. *JAMA* 305 (6):569-575, 2011. doi: 10.1001/jama.2011.90
4. [Krag DN, Anderson SJ, Julian TB, et al](#): Sentinel-lymph-node resection compared with conventional axillary-lymph-node dissection in clinically node-negative patients with breast cancer:

overall survival findings from the NSABP B-32 randomised phase 3 trial. *Lancet Oncol* 11(10):927-933, 2010. doi:10.1016/S1470-2045(10)70207-2

5. [National Lymphedema Network Medical Advisory Committee](#): Position Statement Paper: Lymphedema Risk Reduction Practices. May 2010. Accessed May 8, 2023.

6. [Hughes KS, Schnaper LA, Berry D, et al](#): Lumpectomy plus tamoxifen with or without irradiation in women 70 years of age or older with early breast cancer. *N Engl J Med* 351 (10):971-977, 2004. doi:10.1056/NEJMoa040587

7. [Fisher B, Anderson S, Tan-Chiu E, et al](#): Tamoxifen and chemotherapy for axillary node-negative, estrogen receptor-negative breast cancer: findings from National Surgical Adjuvant Breast and Bowel Project B-23. *J Clin Oncol* 19(4):931-942, 2001. doi:10.1200/JCO.2001.19.4.931

8. [Early Breast Cancer Trialists' Collaborative Group \(EBCTCG\)](#): Effects of chemotherapy and hormonal therapy for early breast cancer on recurrence and 15-year survival: an overview of the randomised trials. *Lancet* 365(9472):1687-1717, 2005. doi:10.1016/S0140-6736(05)66544-0

9. [Davies C, Pan H, Godwin J, et al](#): Long-term effects of continuing adjuvant tamoxifen to 10 years versus stopping at 5 years after diagnosis of oestrogen receptor-positive breast cancer: ATLAS, a randomised trial [published correction appears in *Lancet*. 2013 Mar 9;381(9869):804] [published correction appears in *Lancet*. 2017 May 13;389(10082):1884]. *Lancet* 381(9869):805-816, 2013. doi:10.1016/S0140-6736(12)61963-1

10. [Rosen PP, Saigo PE, Braun DW Jr, et al](#): Predictors of recurrence in stage I (T1N0M0) breast carcinoma. *Ann Surg* 193(1):15-25, 1981. doi:10.1097/00000658-198101000-00003

11. [Swain SM, Baselga J, Kim SB, et al](#): Pertuzumab, trastuzumab, and docetaxel in HER2-positive metastatic breast cancer. *N Engl J Med* 372 (8):724-734, 2015. doi: 10.1056/NEJMoa1413513

## Prognosis for Breast Cancer

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Long-term prognosis depends on tumor stage. Nodal status (including number and location of nodes) correlates with disease-free and overall survival better than any other prognostic factor.

The 5-year survival rate (from the National Cancer Institute's [Surveillance, Epidemiology, and End Results \(SEER\) Program](#)) depends on cancer stage:

- Localized (confined to primary site): 99.0%
- Regional (confined to regional lymph nodes): 85.8%
- Distant (metastasized): 29.0%
- Unknown: 57.8%

**Poorer prognosis** is associated with the following other factors:

- **Young age:** Prognosis appears worse for patients diagnosed with breast cancer during their 20s and 30s than for patients diagnosed during middle age.
- **Race:** Breast cancer death rates from 2015 to 2019 were higher in the United States in non-Hispanic Black females (28 per 100,000) than in non-Hispanic White females (19.9 per

100,000 [1]). Black women are diagnosed at a younger age compared with White women (median 60 versus 63 years) and are more likely to have triple negative disease (2).

- **Larger primary tumor:** Larger tumors are more likely to be node-positive, but they also confer a worse prognosis independent of nodal status.
- **High-grade tumor:** Patients with poorly differentiated tumors have a worse prognosis.
- **Absence of estrogen and progesterone receptors:** Patients with ER+ tumors have a somewhat better prognosis and are more likely to benefit from endocrine therapy. Patients with progesterone receptors on a tumor may also have a better prognosis. Patients with both estrogen and progesterone receptors on a tumor may have a better prognosis than those who have only one of these receptors, but this benefit is not clear.
- **Presence of HER2 protein:** When the *HER2* gene (*HER2/neu* [*erb-b2*]) is amplified, HER2 is overexpressed, increasing cell growth and reproduction and often resulting in more aggressive tumor cells. Overexpression of HER2 is an independent risk factor for a poor prognosis; it may also be associated with high histologic grade, ER- tumors, greater proliferation, and larger tumor size, which are all poor prognostic factors.
- **Presence of *BRCA* gene mutations:** For any given stage, patients with the *BRCA1* gene mutation appear to have a worse prognosis than those with sporadic tumors, perhaps because they have a higher proportion of high-grade, hormone receptor-negative cancers. Patients with the *BRCA2* gene mutation probably have the same prognosis as those without the mutation if the tumors have similar characteristics. With either gene mutation, risk of a 2nd cancer in remaining breast tissue is increased (to perhaps as high as 40%).

## End-of-life issues

For patients with metastatic breast cancer, quality of life may deteriorate, and the chances that further treatment will prolong life may be small. Palliation may eventually become more important than prolongation of life.

Cancer [pain](#) can be adequately controlled with appropriate medications, including [opioid analgesics](#). Other symptoms (eg, constipation, difficulty breathing, nausea) should also be treated.

Psychologic and spiritual counseling should be offered.

Patients with metastatic breast cancer should be encouraged to prepare [advance directives](#), indicating the type of care they desire in case they are no longer able to make such decisions.

## Prognosis references

1. [American Cancer Society](#): Cancer Facts & Figures for African American/Black People 2022-2024. Accessed May 4, 2023.
2. [American Cancer Society](#): Key Statistics for Breast Cancer. Accessed May 4, 2023.

## Key Points

- Breast cancer is the second leading cause of cancer death in women; cumulative risk of developing breast cancer by age 95 is 12%.
- Factors that greatly increase risk include breast cancer in close relatives (particularly if a *BRCA* gene mutation is present), atypical ductal or lobular hyperplasia, lobular carcinoma in situ, and significant exposure to chest radiation therapy before age 30.
- Factors suggesting a poorer prognosis include younger age, absence of estrogen and progesterone receptors, and presence of HER2 protein or *BRCA* gene mutations.
- For most women, treatment requires surgical removal, lymph node sampling, systemic therapy (endocrine therapy or chemotherapy), and radiation therapy.
- Treat with endocrine therapy (eg, tamoxifen, aromatase inhibitor) if tumors have hormone receptors.
- Consider treating metastatic disease to relieve symptoms (eg, with chemotherapy, endocrine therapy, or, for bone metastases, radiation therapy or bisphosphonates), even though survival is unlikely to be prolonged.

## More Information

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The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of these resources.

[NCCN Clinical Practice Guideline](#): Breast Cancer: The National Comprehensive Cancer Network provides guidelines for the diagnosis, staging, and treatment of breast cancer (and other cancers).

[U. S. Preventive Services Task Force](#): Breast Cancer: Medication Use to Reduce Risk: This web site provides the rationale of using medications to reduce the risk of breast cancer in women at high risk and describes the risks of using these medications.

[National Cancer Institute](#): Breast Cancer: This web site discusses the genetics of breast and gynecologic cancers and the screening for and prevention and treatment of breast cancer. It also includes evidence-based information about supportive and palliative care.



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